Indicator 5.3.3: Political Party Engagement with Health and Climate Change

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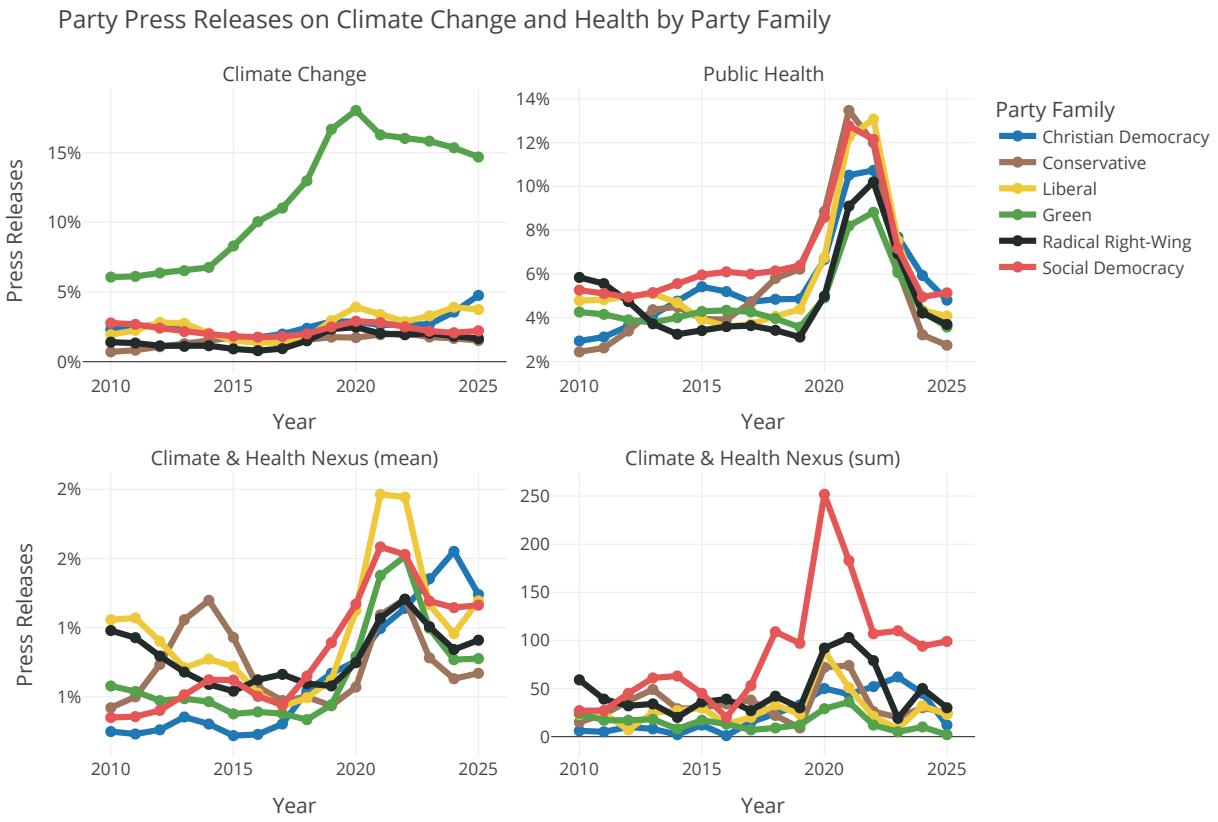
Main Text:

A key obstacle to consolidating political commitment to address the climate crisis remains electing politicians and parties who prioritise climate change and its associated risks to public health. This indicator tracks European political parties' emphasis of public health and climate change in official party communication via press releases. The indicator is based on approximately 571,000 press releases from 127 political parties in 27 European countries and the UK, covering a period from 2010 to 2025. The data were analysed using a fine-tuned language model trained and validated to identify the primary substantive focus of the press releases. The results are presented according to the associated party family of each party. The figure highlights the dramatic influence that the COVID-19 pandemic had on the political agenda, with a significant increase in press releases addressing public health in 2020 and 2021. However, the data also show that press releases focusing on climate change has remained relatively stable over the past 15 years, with Green parties a notable exception. Social Democratic, Liberal, and Agrarian parties have historically been the most likely to link public health to climate change, while radical right-wing parties are the least likely to do so. At the height of the pandemic in 2020, Social Democratic parties issued the greatest number of press releases linking climate change and public health, with notable increases across all party families. These numbers have since declined, with parties focusing on the climate-health nexus in around 0.3 to 1.5 percent of their press releases in 2025.


Party Press Releases on Climate Change and Health by Party Family
Climate Change
Public Health
Climate & Health Nexus (mean)
Climate & Health Nexus (sum)
Press Releases
Year
Party Family
Christian Democracy
Conservative
Liberal
Green
Radical Right-Wing
Social Democracy

Appendix

Indicator 5.3.3

Political party engagement with health and climate change

This indicator tracks European political parties' emphasis of public health and climate change in official party communication via press releases. The indicator is based on approximately 571,000 press releases from 127 political parties in 27 European countries and the UK, covering a period from 2010 to 2025. The data were analysed using a validated fine-tuned language model trained on pre-labeled data that identifies the primary substantive focus of the press releases. The results are presented according to the party family according to ParlGov. The figure highlights the dramatic influence that the COVID-19 pandemic had on the political agenda, with a significant increase in press releases addressing public health in 2020 and 2021. However, the data also show that press releases focusing on climate change has remained relatively stable over the past 15 years, with Green parties a notable exception. Social Democratic, Liberal, and Agrarian parties have historically been the most likely to link public health to climate change, while radical right-wing parties are the least likely to do so. At the height of the pandemic in 2020, Social Democratic parties issued the greatest number of press releases linking climate change and public health, with notable increases across all party families. However, these numbers have since declined, with parties focusing on the climate-health nexus in around 0.3 to 1.5 percent of their press releases in 2025.

Geographic coverage of Europe

This indicator covers 27 European countries and the UK. The 24 EU member states included are: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Lithuania, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden. Additionally, the indicator covers three non-EU European countries: Iceland and Norway (EEA members) and Switzerland, as well as the United Kingdom. The countries were selected based on the availability of press releases from political parties.

Methods

The data were analysed using a fine-tuned language model trained on pre-labeled data that identifies the primary substantive focus of the press releases. The model is trained to predict the substantive focus of the press releases based on the text of the press release. The classification scheme is based on the Comparative Agendas Project (CAP) (Walgrave and Boydstun, 2019) which includes the following categories: defense, technology, culture, macroeconomics, domestic commerce, immigration, healthcare, government operations, housing, social welfare, education, agriculture, environment and climate, foreign trade, civil rights, energy, international affairs, labor, law and crime, transportation, and public lands. One small modification was made to change the focus of the category "environment" to "environment and climate" to better reflect the focus of the indicator.

The model was trained on a dataset of 121,473 examples drawn exclusively from human-labeled and CAP-coded sources, comprising three complementary datasets: (1) 2,122 hand-coded press releases from the PartyPress project (Erfort et al., 2023), double-verified by the research team; (2) 1,208 hand-coded press releases coded with high confidence by research assistants; and (3) 118,147 publicly available CAP-coded press releases from the Comparative Agendas Project (Walgrave and Boydstun, 2019). The combined dataset was split into training (70%), validation (15%), and test (15%) sets using a stratified split by label, yielding 85,031 training, 18,221 validation, and 18,221 test examples. To address class imbalance across the 21 CAP categories, inverse-frequency class weights were applied in the cross-entropy loss function, combined with label smoothing ($\epsilon = 0.1$) to improve model calibration. Training used a cosine learning-rate schedule and early stopping (patience of 4 epochs) to prevent overfitting. A multilingual XLM-RoBERTa-large model pre-trained by Facebook AI (Conneau et al., 2019) was used as the base model for the fine-tuning process. The model achieved a weighted F1 score of 0.78 on a held-out test set of 18,221 examples, indicating strong accuracy in predicting the substantive focus of multilingual press releases. Full results of the model evaluation are presented below. The model has been made publicly available on HuggingFace at: https://huggingface.co/z-dickson/xlm-roberta-large-political-issues.

Table 1: Classification Report for Fine-tuned XLM-RoBERTa-large Model (Test set: $n = 18{,}221$)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Macroeconomics	0.78	0.77	0.78	965
Civil Rights	0.59	0.67	0.63	679
Health	0.84	0.86	0.85	888
Agriculture	0.81	0.82	0.81	352
Labor	0.75	0.79	0.77	585
Education	0.88	0.88	0.88	551
Environment & Climate Change	0.70	0.75	0.72	355
Energy	0.78	0.85	0.81	377
Immigration	0.79	0.84	0.81	287
Transportation	0.85	0.87	0.86	817
Law and Crime	0.80	0.74	0.77	2565
Social Welfare	0.74	0.80	0.76	363
Housing	0.77	0.80	0.78	398
Domestic Commerce	0.68	0.71	0.70	944
Defense	0.78	0.77	0.77	1574
Technology	0.74	0.76	0.75	528
Foreign Trade	0.54	0.65	0.59	142
International Affairs	0.78	0.77	0.77	1876
Government Operations	0.84	0.77	0.80	2925
Public Lands	0.68	0.78	0.73	240
Culture	0.70	0.79	0.74	810
Accuracy			**0.78**	18221
Macro Avg	0.75	0.78	0.77	18221
Weighted Avg	0.78	0.78	0.78	18221

After the model was trained and evaluated, it was used to classify each of the press releases in the dataset. These classifications were then used to create a time series of focus on public health and climate change in the press releases. The time series was created by aggregating the annual number of press releases devoted to climate change, public health, and the intersection between climate change and public health. The time series was then visualized using a line graph, with the x-axis representing the year and the y-axis representing the number of press releases in each category. The results are presented according to the party family according to ParlGov (Döring and Manow, 2012).

Data sources

All data were collected from the official websites of the political parties and included press releases from 2010 to 2025. Data collected for the indicator builds upon previous work in Erfort et al. (2023) and Dickson and Hobolt (2024). Data were collected using a web scraping tool that automatically downloaded the press releases from the official websites of the political parties. The data were then cleaned and pre-processed to remove any irrelevant information, such as HTML tags and formatting. The final dataset included 570,766 press releases from 127 political parties across 27 European countries and the UK.

Caveats/Limitations

The indicator is based on press releases from political parties, which may not accurately reflect the full range of political discourse on health and climate change. The data are also limited to the countries included in the analysis, which may not be representative of the entire European Union. Additionally, the model used to classify the press releases is not perfect and may misclassify some press releases. The model was trained on a dataset of 121,473 examples, but it is possible that some press releases were misclassified due to the complexity of the language used in the press releases. Finally, there are differences in the ways in which political parties utilize press releases, which may affect the results. For example, some parties may issue more press releases than others, or may use different language to describe similar issues. This may lead to differences in the number of press releases issued by different parties, which may affect the results of the analysis.

Future form of the indicator

The indicator will be updated annually to include new data and to reflect changes in the political landscape. The model used to classify the press releases will also be updated periodically to improve its accuracy and to reflect changes in the language used in the press releases. The indicator will also be expanded to include additional countries and political parties, as well as additional data sources, such as social media and news articles. This will allow for a more comprehensive analysis of the political discourse on health and climate change in Europe.

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Additional analysis

The indicator will also be expanded to include additional analyses that capture party positions on health and climate change. These future analyses will be used to assess the party's positions on health and climate change, as well as the party's overall engagement with these issues. The results of these analyses will be presented in future reports and will be used to inform policy recommendations for improving political engagement with health and climate change in Europe.

Supplementary figures

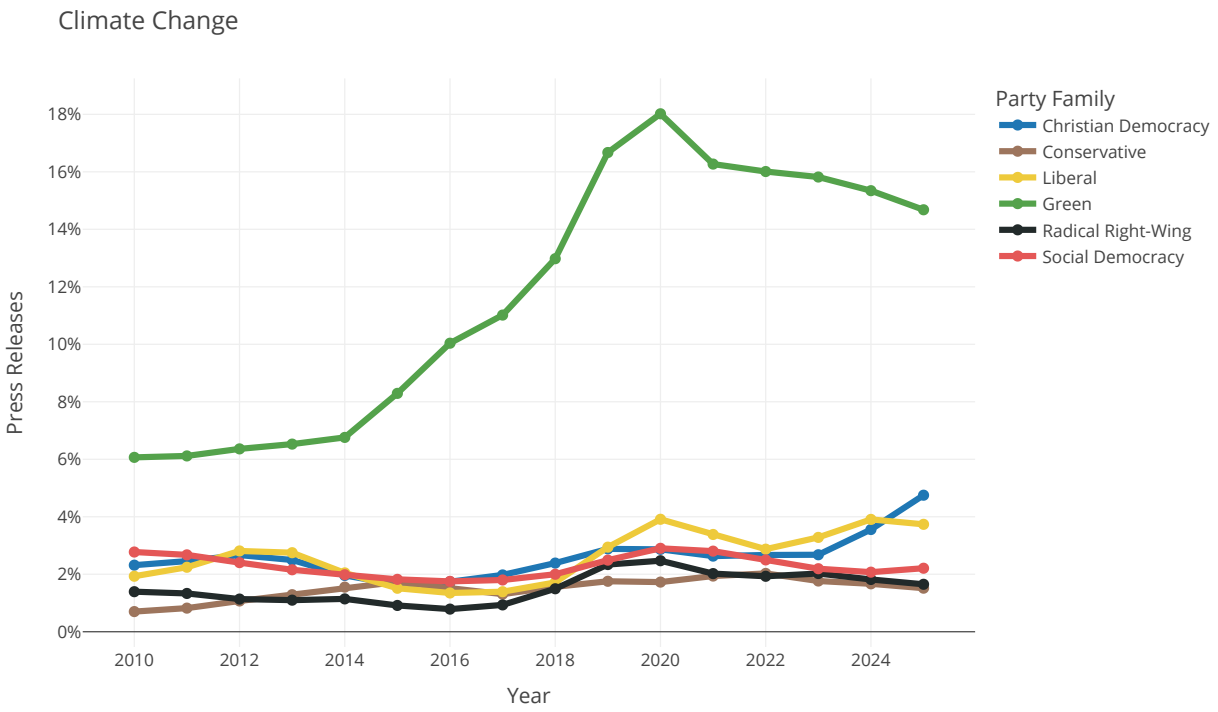

Figure 1: Climate Change: Share of press releases focused on climate change by party family, 2010–2025.

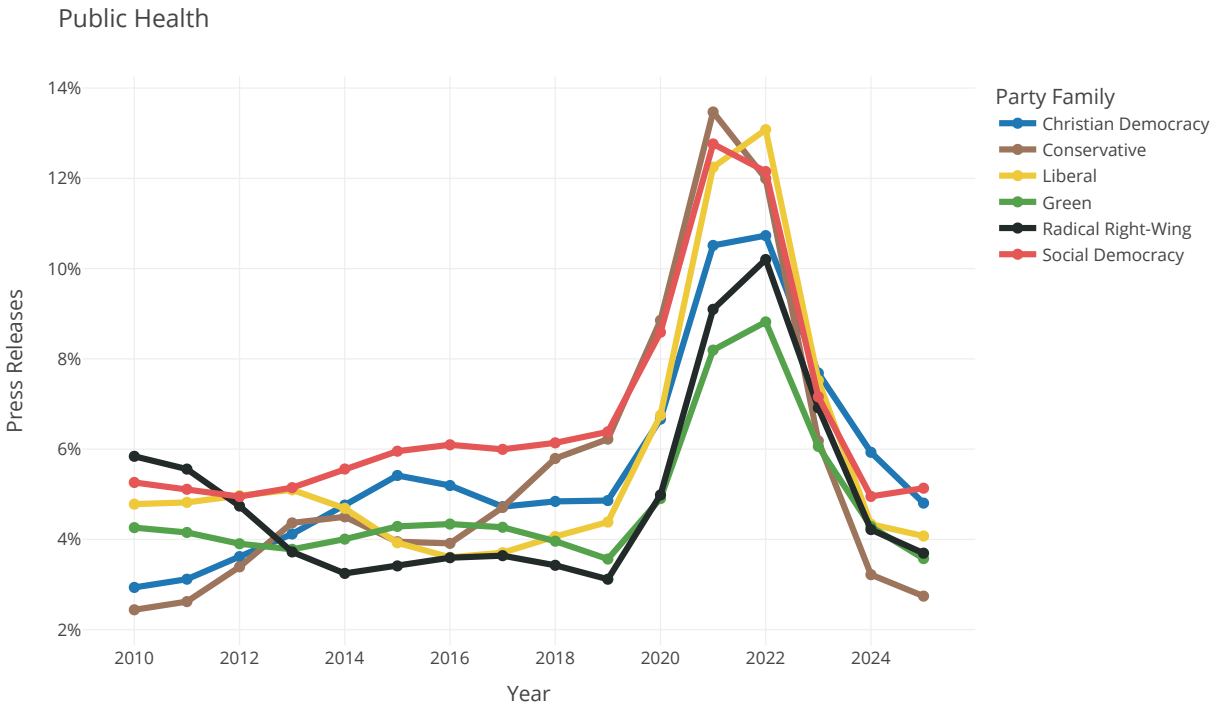

Figure 2: Public Health: Share of press releases focused on public health by party family, 2010–2025.

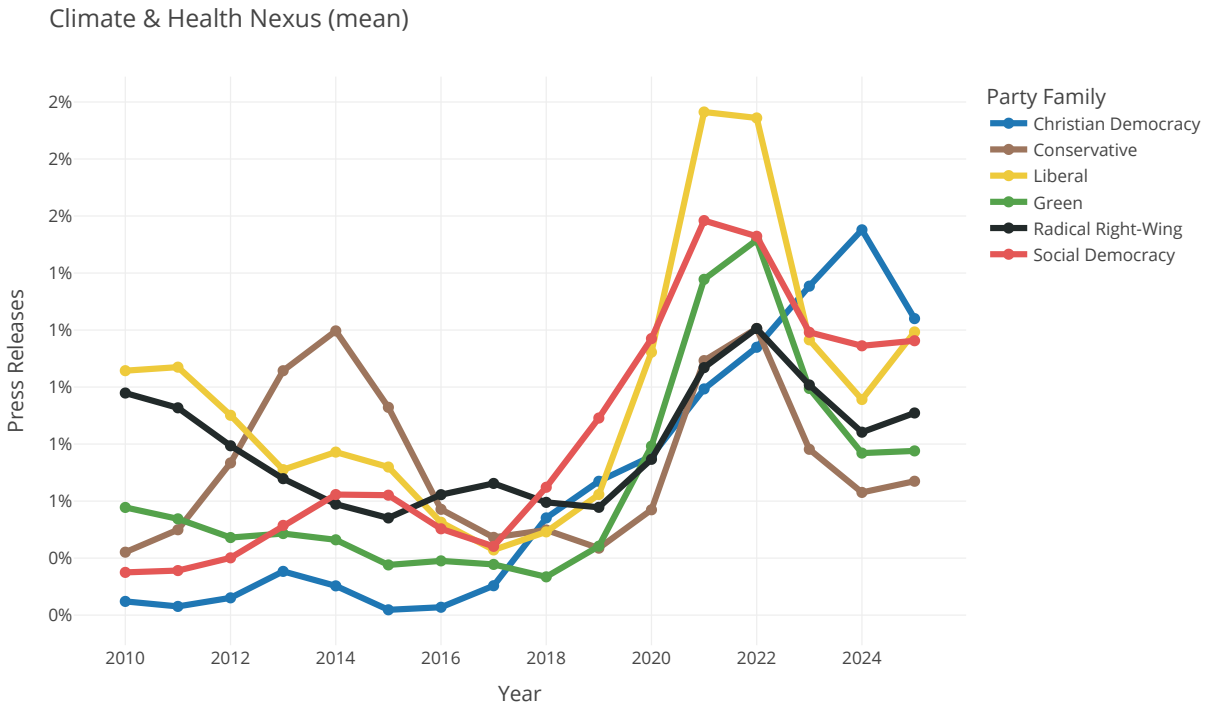

Figure 3: Climate & Health Nexus (mean): Mean share of press releases addressing both climate change and public health by party family, 2010–2025.

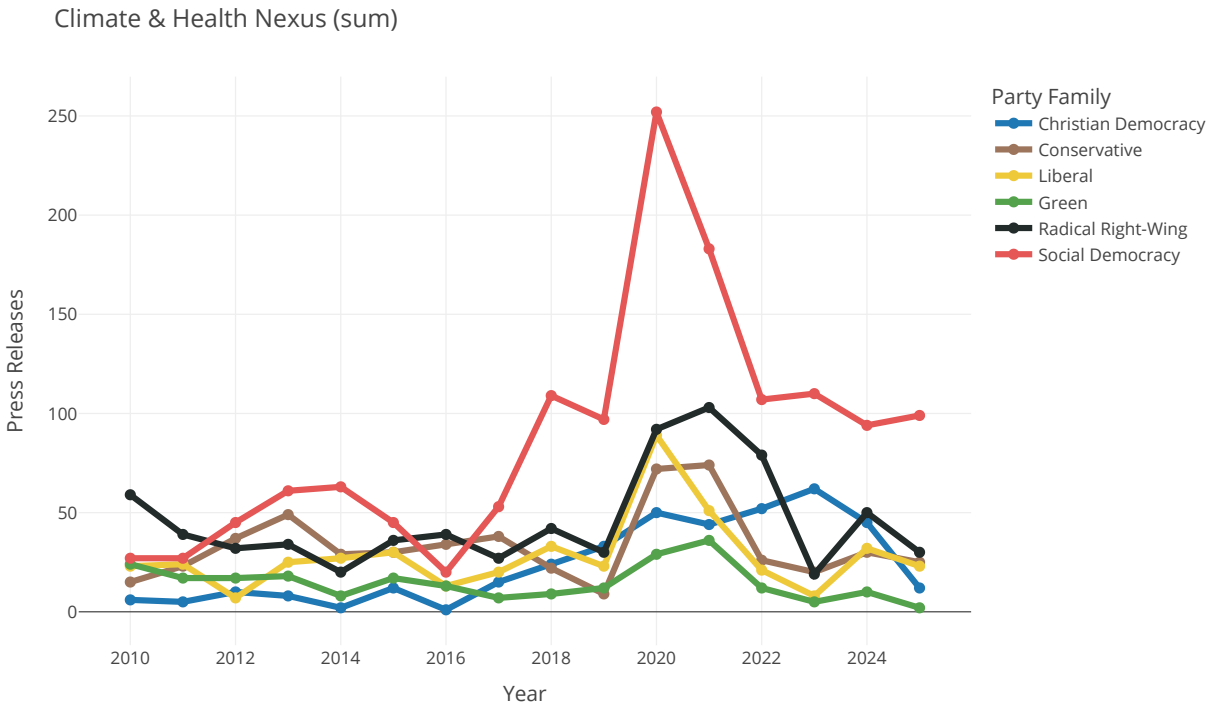

Figure 4: Climate & Health Nexus (sum): Total number of press releases addressing both climate change and public health by party family, 2010–2025.

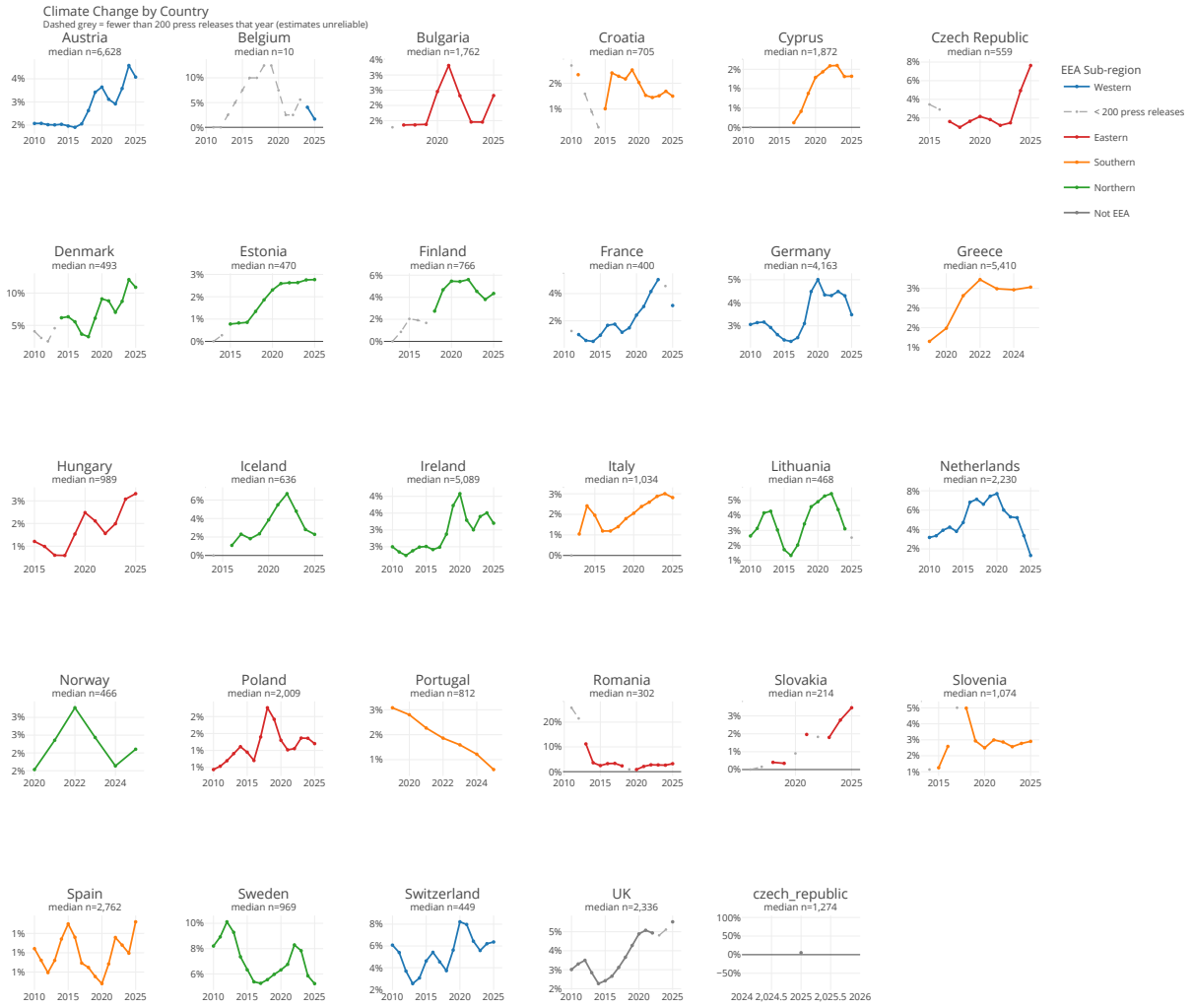

Figure 5: Climate Change by Country: Share of press releases focused on climate change by country, 2010–2025. Countries are coloured by EEA sub-region (Western / Eastern / Northern / Southern). Dashed grey lines indicate years with fewer than 200 press releases, where proportion estimates are unreliable.

Figure 6: Public Health by Country: Share of press releases focused on public health by country, 2010–2025. Countries are coloured by EEA sub-region. Dashed grey lines indicate years with fewer than 200 press releases.

Figure 7: Climate & Health Nexus (mean) by Country: Mean share of press releases addressing both climate change and public health by country, 2010–2025. Countries are coloured by EEA sub-region. Dashed grey lines indicate years with fewer than 200 press releases.

Figure 8: Climate & Health Nexus (sum) by Country: Total number of press releases addressing both climate change and public health by country, 2010–2025. Countries are coloured by EEA sub-region. Dashed grey lines indicate years with fewer than 200 press releases.

References

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2019). Unsupervised cross-lingual representation learning at scale. *CoRR*, abs/1911.02116.

Dickson, Z. P. and Hobolt, S. B. (2024). Going against the grain: Climate change as a wedge issue for the radical right. *Comparative Political Studies*.

Döring, H. and Manow, P. (2012). Parliament and government composition database (parlgov). *An infrastructure for empirical information on parties, elections and governments in modern democracies. Version*, 12(10).

Erfort, C., Stoetzer, L. F., and Klüver, H. (2023). The partypress database: A new comparative database of parties' press releases. *Research & Politics*, 10(3).

Walgrave, S. and Boydstun, A. E. (2019). The comparative agendas project. *Comparative Policy Agendas: Theory, Tools, Data*, page 35.